

# DBMS-LAB(06)

## Question 1:

1.

```
mysql> SELECT emp_name,doj
-> FROM employee
-> WHERE emp_dept IN(
->     SELECT emp_dept
->     FROM employee
->     WHERE emp_name='mark'
-> ) AND emp_name!='mark';
```

| emp_name | doj        |
|----------|------------|
| linklon  | 2008-08-08 |
| linkoln  | 2000-01-01 |

2 rows in set (0.01 sec)

2.

```
mysql> SELECT emp_id,emp_name
-> FROM employee
-> WHERE income >(
->     SELECT AVG(income) as avgsalary
->     FROM employee
-> )
-> ORDER BY income DESC;
```

| emp_id | emp_name |
|--------|----------|
| 2514   | pasin    |
| 2513   | manas    |
| 2508   | obama    |
| 2518   | obama    |
| 2512   | mac      |
| 2522   | mac      |

6 rows in set (0.00 sec)

```
mysql> |
```

3.

```
mysql> SELECT emp_id,emp_name
-> FROM employee
-> WHERE emp_dept IN(
->     SELECT emp_dept
->     FROM employee
->     WHERE emp_name LIKE '%n%'
-> );
```

| emp_id | emp_name |
|--------|----------|
| 2505   | peter    |
| 2506   | mark     |
| 2507   | donald   |
| 2509   | linklon  |
| 2510   | kane     |
| 2512   | mac      |
| 2513   | manas    |
| 2514   | pasin    |
| 2515   | peter    |
| 2516   | mark     |
| 2517   | donald   |
| 2519   | linkoIn  |
| 2520   | kane     |
| 2522   | mac      |

14 rows in set (0.00 sec)

4.

```
mysql> SELECT emp_name,emp_dept,emp_id
-> FROM employee
-> WHERE emp_name IN(
->     SELECT emp_name
->     FROM employee
->     WHERE palce='india'
-> );
```

| emp_name | emp_dept | emp_id |
|----------|----------|--------|
| manas    | accounts | 2513   |
| pasin    | accounts | 2514   |

2 rows in set (0.00 sec)

5.

```
mysql> SELECT max(income)
-> FROM employee
-> WHERE income <(
->     SELECT max(income)
->     FROM employee
-> );
```

| max(income) |
|-------------|
| 600000      |

```
1 row in set (0.00 sec)

mysql> |
```

## Question 2:

1.

```
mysql> SELECT*
-> FROM sailors
-> WHERE sid IN (
->     SELECT sid
->     FROM Reserves
->     WHERE bid=101
-> )
-> ;
```

| sid | sname   | rating | age |
|-----|---------|--------|-----|
| 22  | Dustin  | 7      | 45  |
| 64  | Horatio | 7      | 35  |

```
2 rows in set (0.00 sec)
```

2.

```
mysql> SELECT bname
-> FROM Boats
-> WHERE bid IN(
->     SELECT bid
->     FROM Reserves
->     WHERE sid IN(
->         SELECT sid
->         FROM sailors
->         WHERE sname='Bob'
->     )
-> )
-> ;
```

```
Empty set (0.00 sec)

mysql> |
```

3.

```
mysql> SELECT sname
-> FROM sailors
-> WHERE sid IN(
->     SELECT sid
->     FROM Reserves
->     WHERE bid IN(
->     SELECT bid
->     FROM Boats
->     WHERE color='red'
-> )
-> )
-> ORDER BY age;
```

| sname   |
|---------|
| Horatio |
| Dustin  |
| Lubber  |

3 rows in set (0.00 sec)

4.

```
mysql> SELECT sname,sid
-> FROM sailors
-> WHERE sid IN(
->     SELECT sid
->     FROM Reserves
->     GROUP BY sid,day
->     HAVING COUNT(DISTINCT(bid))>=2)
-> );
```

| sname   | sid |
|---------|-----|
| Dustin  | 22  |
| Lubber  | 31  |
| Horatio | 64  |
| Horatio | 74  |

4 rows in set (0.00 sec)

mysql> |

5.

```
mysql> SELECT sname,sid
-> FROM sailors
-> WHERE sid IN(
->   SELECT sid
->
->   FROM Reserves
->   GROUP BY sid,day
->   HAVING COUNT(DISTINCT(bid))>=2)
-> );
```

| sname   | sid |
|---------|-----|
| Dustin  | 22  |
| Lubber  | 31  |
| Horatio | 64  |
| Horatio | 74  |

4 rows in set (0.00 sec)

```
mysql> |
```

6.

```
mysql> SELECT sid
-> FROM sailors
-> WHERE sid IN(
->   SELECT sid
->   FROM Reserves
->   WHERE bid IN(
->     SELECT bid
->     FROM Boats
->     WHERE color='red' OR color='green'
->   )
-> );
```

| sid |
|-----|
| 22  |
| 31  |
| 64  |
| 74  |

4 rows in set (0.00 sec)

7.

```
mysql> SELECT sname, Age
-> FROM sailors
-> WHERE Age IN(
->     SELECT min(Age)
->     FROM sailors
-> );
```

| sname | Age |
|-------|-----|
| Zorba | 16  |

1 row in set (0.00 sec)

8.

```
mysql> SELECT (COUNT(DISTINCT sname))
-> FROM sailors;
```

| (COUNT(DISTINCT sname)) |
|-------------------------|
| 9                       |

1 row in set (0.00 sec)

9.

```
mysql> SELECT Rating, avg(Age)
-> FROM sailors
-> GROUP BY Rating;
```

| Rating | avg(Age) |
|--------|----------|
| 7      | 40       |
| 1      | 33       |
| 8      | 40.5     |
| 10     | 25.5     |
| 9      | 40       |
| 3      | 44.5     |

6 rows in set (0.00 sec)

10.

```
mysql> SELECT Rating ,avg(Age)
-> FROM sailors
-> GROUP BY Rating
-> HAVING COUNT(sid)>=2;
```

| Rating | avg(Age) |
|--------|----------|
| 7      | 40       |
| 8      | 40.5     |
| 10     | 25.5     |
| 3      | 44.5     |

```
4 rows in set (0.00 sec)
```